

🎓 EDUCATION

B.S. in Physics & Civil Engineering, National Taiwan University

2020-2025

- Obtained dual bachelor's degrees in Physics and Civil Engineering.

M.S. in Structural Engineering, National Taiwan University

2025-2026

- Thesis: **Secondary Weakening in dm-Scale Laboratory Earthquakes: Frictional Origin or Rigid-Body Artifact?**

🔬 RESEARCH EXPERIENCES

Validation of Seismic Parameters via Laboratory Seismic Experiments

2024–Present

Division of Structural Engineering, Department of Civil Engineering, NTU

Advisor: Dr. Chun-Yu Ke

- Conduct dm-Scale laboratory earthquake experiments to investigate earthquake energy budgets.
- Conduct **Finite Element** & **Spectral Boundary Integral** simulation to estimate the energy release rate in different configurations of seismic experiments under the **LEFM** assumption.
- Processed and interpreted simulations using C++ and Python on GPUs, ensuring high accuracy and economically.

Seismic Structural Health Monitoring and Analysis

2023–Present

Institute of Earth Sciences, Academia Sinica

Advisors: Dr. Kuo-Fong Ma, Dr. Wen-Tzong Liang

- Investigated how seismic events influence the dynamic response and structural integrity of buildings.
- Utilized eigenfrequency analysis, spectral analysis, and deconvolution techniques to detect shifts in building health indicators over time.
- Submitted to Peer-Reviewed Journal Article: XXXX , volXX BSSA, 202X

AI-Based Seismic Signal Classification

2023–2025

Institute of Earth Sciences, Academia Sinica

Advisors: Dr. Kuo-Fong Ma, Dr. Wen-Tzong Liang

- Trained CNN-based and a Transformer-based deep learning model to distinguish P-waves, S-waves, and non-seismic signals in real-time.
- Optimized the model for deployment on a Raspberry Pi 4B, enabling low-latency, server-independent earthquake detection on a cost-effective edge device.

Study on Air Drag Force Using High-Speed Camera and Computer Vision

Autumn 2021

Department of Physics, NTU

Advisor: Dr. Ying-Jer Kao

- Employed a high-speed camera to record the free-fall trajectory of a spherical object of different sizes.
- Applied computer vision techniques to track the object's position frame-by-frame.

N-Body Problem Simulation Research

Autumn 2024

Department of Physics & Department of Geosciences, NTU

Advisor: Dr. Ying-Jer Kao, Dr. Yih-Min Wu

- Implemented an N-body simulation in Python, C++, and Fortran.
- Analyzed conservation properties of total energy and angular momentum over time to validate the system's fidelity.
- Integrated the Barnes-Hut algorithm for hierarchical decomposition, significantly improving computational efficiency.

HONORS

Taiwan Farmers' Association Scholarship

Conferred to one of the top 300 performing undergraduate students nationwide.

Autumn 2023

NTU Alumni Excellence Scholarship

Awarded for exceptional academic achievement at NTU, with a GPA of 4.23/4.3.

Summer 2024

Taiwan Farmers' Association Scholarship

Conferred again to one of the top 300 performing undergraduate students nationwide.

Autumn 2024

Taiwan Farmers' Association Scholarship

Conferred to one of the top 300 performing graduate students nationwide.

Autumn 2025

PUBLICATIONS

Peer-Reviewed Journal Publications

- [1] Chun-Yu Ke, **Gauss T. Chang**, Gregory C. McLaskey, David S. Kammer, Chris Marone. (2026) Nonlocal Elastic Unloading as a Mechanism for Breakdown Work Scaling in Laboratory Earthquakes. *Earth and Planetary Science Letters*, doi: [10.1016/j.epsl.2026.120283](https://doi.org/10.1016/j.epsl.2026.120283).
- [2] QGIS: A low-cost Structural Health Monitoring System
- [3] The paper for my FEM.

Selected Conference Abstracts

- [1] **Gauss T. Chang**, Chun-Yu Ke. *Secondary Weakening in dm-Scale Laboratory Earthquakes: Frictional Origin or Rigid-Body Artifact?*. JpGU-AGU Joint Meeting 2026, Chiba, Japan, 24–29 May 2026.
- [2] **Gauss T. Chang**, Chris Marone, Chun-Yu Ke. *Earthquake Stress-Drop Estimation in Laboratory Experiment with Machine Learning*. the Japan Geoscience Union Meeting 2025, Chiba, Japan, 25–30 May 2025.
- [3] **Gauss T. Chang**, Wen-Tzong Liang, Utpal Kumar, Kuo-Fong Ma, Li-Wei Chen. *Orchestrating Structural Safety with QGIS: Vision for Extensible Building Arrays*. Japan Geoscience Union Meeting 2025, Chiba, Japan, 25–30 May 2025.
- [4] Chun-Yu Ke, **Gauss T. Chang**, Gregory McLaskey, Chris Marone. *Earthquake Rupture Speed Dependence on Normal Stress in Laboratory Experiments*. EGU General Assembly 2025, Vienna, Austria, 27 April–2 May 2025.
- [5] **Gauss T. Chang**, Wen-Tzong Liang, Utpal Kumar, Kuo-Fong Ma, Hung-Jun Chou, Li-Wei Chen. *QGIS: A Scalable and Cost-Effective Solution for Advanced Structural Seismic Monitoring*. AGU24 Annual Meeting, Washington, D.C., 9–13 December 2024.
- [6] **Gauss T. Chang**, Chris Marone, Chun-Yu Ke. *Inferring Earthquake Stress Drops with Passive Acoustics in Laboratory Experiments Using Machine Learning*. Taiwan Geosciences Assembly 2025, Nangang, Taipei, Taiwan, 16–19 June 2025.
- [7] **Gauss T. Chang**, Wen-Tzong Liang, Utpal Kumar, Kuo-Fong Ma, Hung-Jun Chou, Li-Wei Chen. *Natural Frequencies of Low-Rise Reinforced Concrete Buildings Determined With QGIS*. Taiwan Geosciences Assembly 2025, Nangang, Taiwan, 16–19 June 2025.
- [8] **Gauss T. Chang**, Wen-Tzong Liang, Hung-Jun Chou, Utpal Kumar, Kuo-Fong Ma. *Toward Structural Health Monitoring with QGIS Network*. TEC24 Annual Meeting, Tainan, Taiwan, 23–24 October 2024.